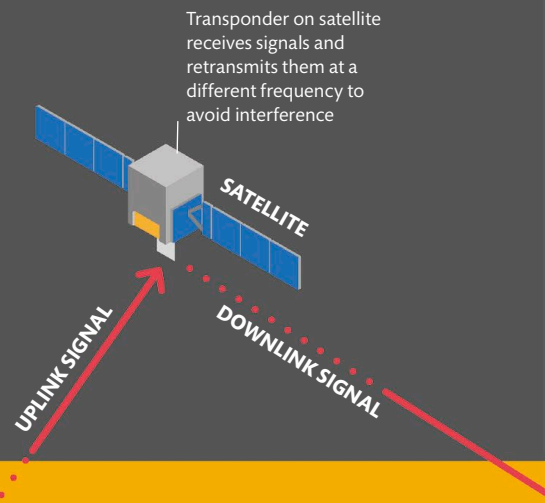
Television broadcasting

Television broadcasting enables video content to reach anyone with a television set. Before appearing on viewers' screens, television programs are transmitted using three kinds of broadcasting technology: terrestrial (using ground-based antennas), satellite, or cable.



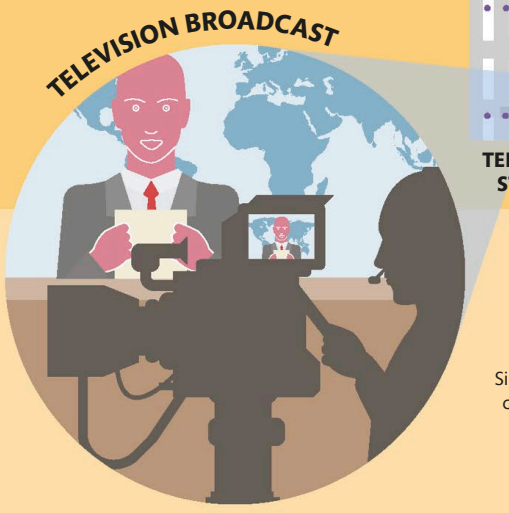
From studio to screen

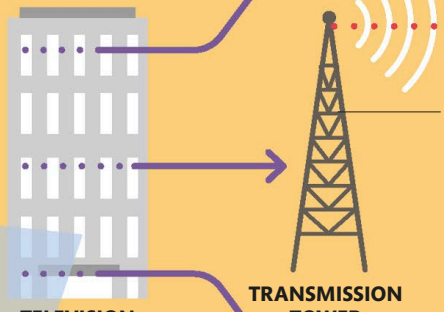
Television scenes are captured with video cameras and microphones, which record visual and audio information as electrical signals. These signals, which contain instructions about exactly how television sets can reconstruct the scene, are modulated (see pp.182–183) and transmitted to viewers' homes via satellite, terrestrial, or cable broadcasting. Each television channel transmits its programs by using signals of different sets of frequencies.



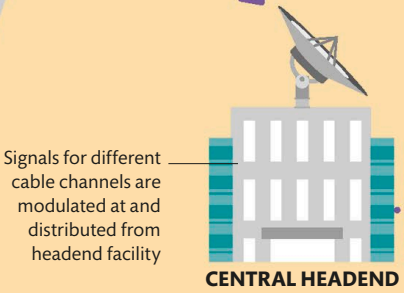
Satellite broadcasting
Satellite television is delivered to homes via a communications satellite, which relays signals in the form of radio waves to viewers' satellite dishes. Satellite television can be accessed even in remote areas and offers more channels than terrestrial broadcasting.

Converting scenes to signals
Modern cameras focus light onto a charge-coupled device, which measures and records the light across each point in a frame. This information—along with recorded sound—is converted into electrical signals ready for transmission.





Terrestrial broadcasting
Terrestrial broadcasting refers to signals transmitted directly from a television station to homes. Terrestrial television was the only available type of television broadcasting until the 1950s.



Cable broadcasting
Cable television is delivered to customers using optical signals transmitted through underground fiber-optic cables (see pp.184–185). The same cables may also be used for Internet access and telephone connections.